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Data Science

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The Science behind Running

"Every morning in Africa, a gazelle wakes up; it knows it must outrun the fastest lion or it will be killed. Every morning in Africa, a lion wakes up; it knows it must run faster than the slowest gazelle, or it will starve. It doesn't matter whether you're the lion or a gazelle—when the sun comes up, you'd better be running (African Proverb)." As world record running performances continue to improve by microscopic margins of a second, the theoretical limit of human achievement approaches a mathematical asymptote. This raises the question: what are the physiological limitations of human endurance? Human endurance is limited by factors such as maximum oxygen uptake (VO2max), blood lactate threshold, running economy, metabolism, neurological factors, genetic makeup, and mathematical modeling.

To kick off, around 300,000 years ago, Homo sapiens derived the capability of endurance running from persistent scavenging and hunting of prey. Physiological features such as sweat glands, Achilles tendons, the compound knee joint, and the gluteus maximus likely evolved from diligent conditioning. The earliest records of competitive running can be dated back to 1829 BCE, with religious festivals organizing events in Greece, Egypt, Asia, and the East African Rift. The Olympics originated from Ancient Greece, taking place from 776 BCE to 393 AD.

"Only the disciplined ones in life are free (Eliud Kipchoge)." The first organized marathon was held in Athens at the 1896 Olympics. The inspiration for the 26.2-mile distance was pioneered by the legend of Pheidippides, an ancient Greek messenger who raced from the site of Marathon to Athens to announce victory over the invading army of Persians in 490 BCE.

After making an act of heroism, the messenger collapsed and died from exhaustion after running nearly 25 miles. Nearly 2,500 years later, Eliud Kipchoge of Kenya did the unthinkable. On October 12, 2019, in Vienna, Austria, Kipchoge broke the two-hour barrier, running a time of 1:59:40.2, solidifying him as the greatest marathoner of all time.

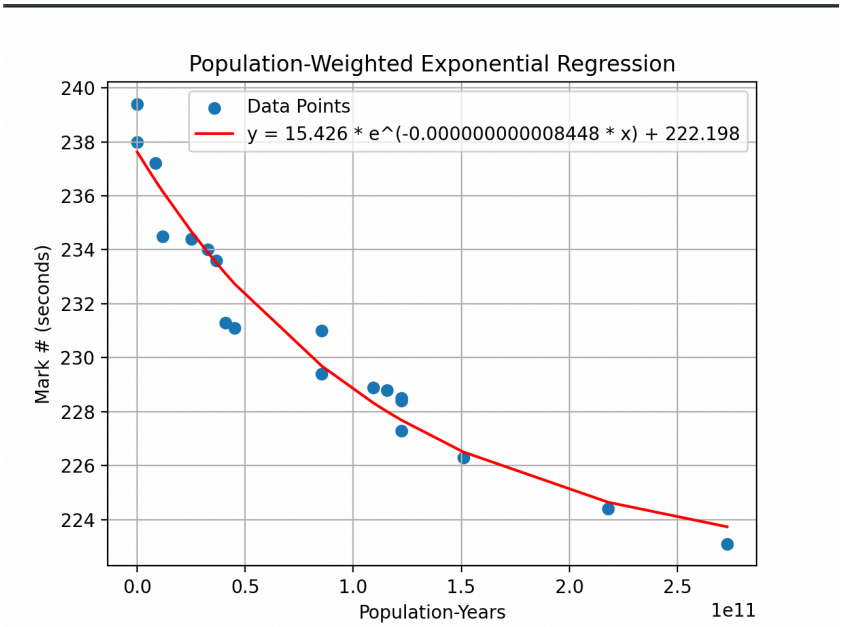
"The man who can drive himself further once the effort gets painful is the man who will win (Sir Roger Banister)." On May 6, 1954, Sir Roger Banister of England, at age 25, ran one mile in 3:59.4, becoming the first person in history to break the four-minute barrier. Since this historic athletic feat, over 1,600 male athletes have broken this barrier, inaugurating this psychological barrier as the international standard of all male elite-distance runners. Hicham El Guerrouj of Morocco, at age 24, set the still-standing world record in 1999 with a time of 3:43.13.

To initiate the investigation, there are two main types of cellular respiration used during exercise: aerobic and anaerobic. Aerobic respiration is physical exercise of low to high intensity that requires molecular oxygen to break down carbohydrates, fatty acids, and amino acids to produce adenosine triphosphate (ATP). Anaerobic respiration is physical exercise of high intensity that requires electron acceptors to oxidize carbohydrates, fatty acids, and amino acids to produce ATP, releasing lactic acid as a byproduct. When it comes to endurance running, aerobic capacity has been shown to be a superior predictor of running performance.

Moreover, having a high VO2max, high blood lactate threshold, and high running economy play key roles in endurance performance. In the current model, VO2max sets the upper limit for aerobic metabolism, while the blood lactate threshold is related to the fraction of VO2max that can be sustained in competitive events. Running economy appears to play a role in

VO2max and blood lactate threshold to determine the actual running speed at lactate buildup. Neurological factors also play important roles in who becomes a champion. These complex physiological factors go far beyond simple explanations and remain an important puzzle for physiological integration well into the future. It should be noted that genetic composition plays a vital role in the attainment of world-class status. Although geographical diversity argues against a simple genetically based gene, there seems to be a correlation of genotypes that produce a more efficient transfer of energy and reduce fatigue.

Understandably, there are many biological variables involved in determining peak human performance. Each must be carefully analyzed to identify potential limits. Looking at the men's mile world record progression from the IAAF era yields an improvement of 31.27 seconds over the last 100 years. Theoretically, the fastest an elite athlete could ever dream of running one mile is limited to the speed of light. In theory, a time faster than 5.337026 × 10^-6 s is impossible. More realistically, if Usain Bolt were to run his 100-meter world record of 9.58 seconds for an entire mile, the time on the clock would read 2:34.18, establishing our lower limit. It is logical to conclude that running a mile at David Rudisha's 800m world record pace of 1:40.91 is impossible, which constitutes a lower limit of 3:21.82.



The goal of this analysis is to determine the asymptote, representing the fastest possible time. Using linear regression, we can plot the world record on the y-axis and the years on the x-axis. According to the graph, there is no asymptote insight, meaning that the linear regression predicts a negative time if extrapolated far enough. The problem with our graph is a lack of data. Plotting population-years, as a function of the elapsed time multiplied by population, gives us an exponential decay with the asymptote yielding a time of 3:42.2, 0.93 seconds short of Hicham El Guerrouj's world record of 3:43.13.

In closing, human physiology has limitations. Elite athletic performance involves the integration of VO2max, blood lactate threshold, running economy, neurological factors, and genetic makeup to function collectively to efficiently transfer energy from aerobic and anaerobic respiration. In the quest for faster times, scientific innovations, technological advances, and pharmaceutical doping may continue to push the boundaries of human performance.

Works Cited

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